Husnain Ali

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PROFILE

AI Research and Development Engineer with around five years of industry experience in natural language processing and computer vision. My interest has moved from building AI applications based on models to understanding the underlying models, the representations and mechanisms they learn and encode, and reverse engineering the algorithms they have learned.

RESEARCH INTERESTS

Mechanistic interpretability • Explainable AI • Energy-based models and JEPA • Non-autoregressive language models (e.g. diffusion LLMs)

RESEARCH & PROFESSIONAL EXPERIENCE

Isgen Ltd **Nov 2025 – Present**
Founding Research Engineer *Manchester, UK (Remote)*

- Designed and trained **multilingual AI-generated-text detection models** that extend coverage beyond English; in production for **10M+ active users**, with **22M+ documents** scanned.
- Devised a **multi-head classification architecture** (multiple classification layers over a shared backbone) that beat single-LoRA and full fine-tuning baselines on accuracy across all supported languages.
- Reached **99.6% AUROC** on multilingual held-out benchmarks: **99.5% recall at a 1% false-positive rate**, and a 0.1% false-positive rate at 95% recall.
- Researched **interpretability methods** to expose which patterns the detector relies on — feature attribution (integrated gradients, SHAP), probing classifiers and attention/activation analysis; also evaluated sparse auto-encoders, which did not improve on these — and shipped the results as **sentence- and phrase-level explanations** shown to users.
- Validated explanations with **faithfulness tests** and product-side signals such as user appeal rates.
- Helped extend detection from text to **AI-generated images**; video detection is in development.

Smodin.io **Aug 2021 – Oct 2025**
AI Research and Development Engineer *Casper, USA (Remote)*

- Led the strategy and development of AI tools for education, including **Omni**, a real-time multilingual assistant for college-level questions that handles **450k+ calls per month** at over 98% accuracy.
- Fine-tuned open-source language models (Mistral, Gemma, Llama-3) for targeted tasks, including **preference-based alignment with PPO and DPO**; ran distributed training on GCP, FluidStack and Vast.ai.
- Researched and prototyped computer vision systems: text-in-image translation, OCR and visual question answering.
- Built NLP and retrieval-augmented generation pipelines (Transformers, sentence-transformers, FAISS, FastAPI) for the law, finance and medical domains.

MathLeaks **Apr 2021 – Jul 2021**
Computer Vision Research and Development Engineer *Sweden (Remote)*

- Surveyed and benchmarked state-of-the-art handwritten mathematical expression recognition models (online and offline) on an in-house dataset.
- Trained and compared several transformer-based architectures, producing a prototype that converts images of handwritten maths into LaTeX.

TEACHING & LEADERSHIP

Google Developer Student Club, NUST **Jan 2021 – Dec 2021**
AI Team Lead *Islamabad, Pakistan*

- Delivered workshops on data science and applied AI, and mentored students one-to-one on applying deep learning to real-world problems.

EDUCATION

National University of Sciences and Technology (NUST) **Sep 2019 – Jun 2023**
Bachelor of Software Engineering *Islamabad, Pakistan*

TECHNICAL SKILLS

Languages: Python, C/C++, Java
ML & research: PyTorch, Hugging Face Transformers, Accelerate, PEFT/LoRA, bitsandbytes, PyTorch Lightning, Keras, scikit-learn, OpenCV, NumPy, pandas, matplotlib
Interpretability: feature attribution (integrated gradients, SHAP), probing classifiers, attention/activation analysis, sparse auto-encoders, faithfulness evaluation
LLMs & retrieval: Fine-tuning (SFT, PPO, DPO), RAG (FAISS, Annoy, Milvus, Pinecone), multi-agent pipelines (LangChain, AutoGen)
Infrastructure: GCP, Azure, Vast.ai, FluidStack, Paperspace, Docker, FastAPI, Git, MLOps